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Personalized learning path optimization based on enhanced deep neural network: higher education teaching model integrating learner behavior and cognitive style

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Abstract

As information technology and artificial intelligence advance rapidly, personalized learning has emerged as a key approach to enhancing teaching quality and learning outcomes in higher education. Current research considers behavior or cognitive style separately, lacks effective integration, and traditional methods are mostly based on static analysis, which makes it difficult to cope with students' dynamic needs and real-time feedback, and lacks flexibility and adaptability. To address the above problems, this paper designs an effective model for optimizing personalized learning paths using Enhanced Deep Neural Networks (EDNN). The model adopts the Actor-Critic framework to introduce a multilayer perceptron (MLP) in the Actor part to fuse behavioral data and cognitive style, and a long short-term memory (LSTM) neural network in the Critic part to process time series data, thereby realizing the selection and dynamic adjustment of personalized learning paths. Experiments show that the model proposed in this study can integrate learners' behavioral data and cognitive styles to optimize personalized learning paths. Compared with the Q-learning model and the Deep Q-learning Network (DQN) model, the model in this study is superior in learning speed, stability, adaptability, etc. During the experiment, the model studied in this paper can reduce the loss to 0.1 after 40 iterations and converge quickly in the later stage. The shortest average feedback response time of 100 experiments is 1.123 s, and the shortest average path adjustment calculation time is 2.010 s. The recommendation effect of different deep neural networks (DNN) on personalized learning paths is analyzed. This model outperforms others in terms of accuracy for personalized learning path recommendations. The model can efficiently integrate learners' behavioral data and cognitive styles, adjust learning paths in real-time, and demonstrate rapid response and optimization capabilities in practical applications.

Keywords Personalized learning path, Enhanced deep neural networks, Actor-critic framework, Behavioral data and cognitive style, Long short-term memory

1 Introduction

As information technology advances rapidly, especially with the widespread adoption of artificial intelligence and big data, personalized learning [1–3] guaranteed higher education teaching quality and student performance. However, the traditional teaching model relies too much on unified course content and progress, and does not consider the differences in students' behavior and cognition. Although many studies have begun to focus on the design of personalized learning paths in recent years [4], most of these studies are limited to static analysis and do not study student growth and data fusion. In addition, existing personalized learning models usually ignore the behavioral data generated by learners during the learning process and their inherent cognitive style differences, resulting in insufficient accuracy and personalization of learning path recommendations. Therefore, an important research area focuses on integrating students' behavioral data with cognitive style data, while dynamically adjusting learning paths to align with students' evolving needs throughout the learning process. This has become a core aspect of personalized education today.

In recent years, with the development of artificial intelligence technology [5, 6], many studies have begun to explore data-driven personalized learning path recommendation methods. Traditional personalized learning path optimization methods mainly recommend learning paths based on students' grades, test results, and basic personal information. Most of these methods simply perform statistics on the data or formulate certain rules to analyze the data. Hwang Gwo-Jen et al. used technologies such as emotion sensors to collect and analyze learners' behavioral data and recommended learning paths based on the analysis results [7]. Kardan Ahmad A. et al. took learners' own growth factors into full consideration and proposed a new adaptive learning path algorithm based on their research to develop personalized learning paths [8]. Smaili El Miloud et al. considered the impact of the environment on learners during the learning process and developed learning paths based on the research conducted by learners in different learning environments [9]. Basuki Ari implemented a course system that can automatically update courseware based on the course sorting problem provided by students [10]. Liu Ziyu et al. mined student learning data and then built a model to analyze the data to recommend personalized, highly similar learning paths for general students [11]. Li Xiaomei et al. used the user's English proficiency, learning preferences and past performance to recommend tailored English listening learning paths for users [12]. Idris Norsham et al. used soft computing technology to deal with the uncertainty and incompleteness of problems, replacing the adaptive method in the adaptive learning system to achieve the customization of personalized paths [13]. Meng Lingling introduced a novel method for generating learning paths, which is based on knowledge structure and learning diagnosis [14]. Iatrellis Omiros et al. configured an environment with a process execution engine including a semantic infrastructure, through which learning paths are configured [15]. Obeng Asare Y. et al. extended the functionality of the Spark framework using ant colony and nearest neighbor techniques and genetic algorithms to implement a complementary learning path personalization architecture [16]. However, these methods usually ignore the differences in students' cognitive styles and learning behaviors, and show low flexibility in dealing with students' dynamic feedback. Therefore, most of these studies are limited to static path recommendations and lack real-time response to students' personalized needs.

In recent years, deep reinforcement learning, as a dynamic decision-making method, has been widely used in personalized recommendation systems. Through reinforcement learning, the system can learn the optimal strategy in the process of interaction between students and the environment (learning content). Li Xiao et al. used the transition simulation estimator for the deep Q-learning algorithm and studied the problem of assuming latent feature persistence in personalized learning path research [17]. Xiong Li et al. integrated Transformer models, generative adversarial networks, and reinforcement learning techniques to enhance personalized learning and optimize its trajectory [18]. Wang Min and Zhihan Lv used deep learning algorithms to effectively classify students based on their learning data and designed diversified customized teaching content based on the performance of the Internet of Things [19]. Su Yingwei and Yuan Wang used reinforcement learning algorithms to explore the intelligent teaching methods of erhu in music, and proposed an intelligent generation model of erhu fingering and an automatic generation algorithm of erhu fingering [20]. Tang Xueying et al. formulated the recommendation strategy problem of personalized recommendation system based on Markov decision framework, and proposed a reinforcement learning method to solve this problem [21]. Pang Guangyao et al. introduced a deep reinforcement learning-based recommendation system utilizing hierarchical attention and sample-enhanced priority experience replay, successfully addressing the cold start issue [22]. Shin Jinnie and Okan Bulut built an intelligent recommendation system using reinforcement learning methods by collecting data such as each student's test time and test scores [23]. To solve the problem that the existing system is difficult to accurately match students' preferred courses, Lim Ducsun et al. proposed a personalized major subject recommendation method based on deep reinforcement learning [24]. To solve the problems of low accuracy and poor recommendation quality in traditional personalized learning resource recommendation methods, Li Zilong et al. proposed an improved method based on deep reinforcement learning [25]. However, most of these methods only focus on behavioral data and ignore the differences in students' cognitive styles, resulting in limited optimization effects on personalized learning paths. In addition, existing methods often rely on static evaluation and are difficult to dynamically adjust during students' learning process.

In this study, two different types of data, behavioral data and cognitive style data, are processed differently. The Actor-Critic architecture [26–28] is used to fuse these two different types of data in the embedding layer of the Actor part of the model to obtain an input that fuses the features of these two types of data. In addition, the MLP network [29, 30] is introduced in the Actor part and activated by the nonlinear function ReLU. The LSTM [31, 32] network model is introduced in the Critic part to process data related to time series such as historical learning and generate Q values in this part to evaluate the current path-state pair. The gradient update strategy [33] is used for the Actor part to maximize the probability of path selection, and the time-division error [34] is used for the Critic part to update the Q value so that the estimated Q value and the target Q value are as close as possible. The experimental results show that the personalized learning path optimization model studied in this paper can effectively integrate behavioral data and cognitive style data to perform personalized learning path selection and dynamically adjust the learning path. The model studied in this paper is compared with the Q-learning [35] model and the DQN36 model. The results show that the model studied in this paper performs well in terms of performance indicators such as learning speed, adaptability and stability. The loss of the model during the convergence process is analyzed. The model can reduce the loss to less than 0.1 after 40 rounds of iterations and converge quickly. The minimum values of the average feedback response time and the average path adjustment calculation time of 100 experiments were 1.123 s and 2.010 s respectively. The results of analyzing different DNN models for personalized learning path recommendation show that the accuracy of personalized learning path recommendation in three different dimensions of behavioral data, cognitive style, and learning history of the model studied in this paper is higher than that of other models. This research suggests that the proposed model offers a practical approach for optimizing and adjusting personalized learning in real-time, and provides important theoretical support and practical significance for improving the quality of education and meeting the personalized learning needs of students.

The key findings of our study include:

(1) The successful integration of learners' behavioral data and cognitive styles to optimize personalized learning paths.
(2) Superior performance in learning speed, stability, and adaptability compared to Q-learning and Deep Q-learning Network (DQN) models.
(3) Rapid response and optimization capabilities in practical applications, with the shortest average feedback response time of 1.123 s and the shortest average path adjustment calculation time of 2.010 s.

2 Data processing

2.1 Original data structure

This paper mainly studies the integration of learners' learning behavior data and cognitive style data. The data structures of these two types of data are explained below.

Table 1 consists of the data structure of the learner behavior data studied in this paper. It is obtained through the interaction data of students on the learning platform. The learner behavior data includes learning time, learning frequency, homework grades, knowledge points mastered, interaction status, and historical learning path selection. There are two main data types for these types of data, continuous numerical type and discrete numerical type, both of which are numerical data types.

Table 2 shows the cognitive style data of learners in this study, which were collected through questionnaires, personalized tests, and learning behavior analysis. The data mainly covers four aspects: learning preferences, information processing style, learning strategies, and thinking styles. These data are all categorical, and specifically, each aspect

Table 1 Behavior data table

Feature name	Data types	Data description
Study time	Continuous numeric type	The length of time learners spend on a learning activity
Learning frequency	Discrete numeric type	The learning frequency of the learner
Homework grades	Continuous numeric type	The learner's score or performance on the learning assessment
Knowledge points mastered	Discrete numeric type	Learners' mastery of different knowledge points
Interaction data	Continuous numeric type	The number of times learners interact with the platform
Learning path selection history	Discrete numeric type	The learning path the learner has chosen

Table 2 Cognitive style data table

Feature name	Data types	Data description
Learning preferences	Category type	Learners' learning style preferences, For example, visual, auditory, kinesthetic, etc
Information processing style	Category type	Learner thinking style, Whether tend to prefer detail analysis, overall understanding, intuitive thinking, etc
Learning strategies	Category type	Learners' learning strategies, such as reflective learning, applied learning, critical learning, etc
Thinking style	Category type	The learner's thinking style type, such as analytical, comprehensive, creative thinking, etc

has different subcategories. For example, learning preferences include visual, auditory, and kinesthetic subcategories.

2.2 Data preprocessing

The data types of learner behavior data and cognitive style data processed in this study are different, so the two types of data are processed separately and then merged.

For the learner behavior data studied in this paper, the data type is continuous numerical type or discrete numerical type. Both are numerical types, so a unified processing method is used. First, data cleaning for learner behavior involved managing missing values and detecting outliers.

This paper studies the way to deal with missing values by using mean filling, that is, the missing behavior data is filled with the mean of the column. The mean formula is:

$$x_{filled} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{1}$$

In formula (1), x_{filled} is the fill value, x_i is the i-th data in the column, and n is the number of samples in the column.

This paper explores an approach for handling outliers using the Z-score method, where the Z-scores of individual data points are calculated, and those exceeding a specific threshold are removed.

The calculation formula of Z-score is:

$$z_i = \frac{x_i - \mu}{\sigma} \tag{2}$$

In formula (2), μ is the mean of the data in this category, σ is the standard deviation of the data in this category, and x_i is the sample value.

After cleaning learner behavior data, dimensional differences between various data categories are eliminated to ensure consistency, this study normalized the behavior data and scaled all data to the [0,1] interval for easy processing.

Normalization formula:

$$x\prime = \frac{x - \min(x)}{\max(x) - \min(x)} \tag{3}$$

In formula (3), min(x) and max(x) represent the minimum and maximum values of the data.

Most of the learner behavior data studied in this paper are time series data. Therefore, while preprocessing the data, some time series features are also extracted to facilitate subsequent analysis of the time series. Due to the needs of subsequent experiments, this paper studies the sliding window features of learner behavior data.

The statistical formula is:

$$mean_t = \frac{1}{n}\sum_{i=t-n+1}^{t} x_i \tag{4}$$

In formula (4), $mean_t$ is the average value of the selected time window, x_i is the time series data point, and n is the size of the selected time window.

Table 3 shows the partial window calculation process table for time series data learning time. The time window size is selected as Table 3.

Since the cognitive style data studied in this paper is categorical data, it cannot be processed together with the above numerical data processing method. In this study, the cognitive style data is first encoded using the One-Hot encoding method. For example, there are three categories of learning preference data under the cognitive style data category: visual, auditory, and kinesthetic. The encoding method for these three categories is: visual: [1,0,0]; auditory: [0,1,0]; kinesthetic: [0,0,1]. Then the expression form of learning preference data is:

$$V_{LP} = [V_1, V_2, V_3] \tag{5}$$

In formula (5), V_1, V_2, V_3, and V_{LP} represent the one-hot encoding of visual, auditory, kinesthetic, and learning preferences, respectively. Similarly, the encoding form of cognitive style data is:

$$V_{CS} = [V_{LP}, V_{IPS}, V_{LS}, V_{TS}] \tag{6}$$

In formula (6), V_{CS}, V_{LP}, V_{IPS}, V_{LS}, and V_{TS} represent the one-hot encoding of cognitive style, learning preference, information processing style, learning strategy, and thinking style, respectively. Different categories of One-Hot encoding have different encoding lengths. Table 4 shows the data of One-Hot encodings of different lengths.

As shown in Table 4, different categories have different coding lengths, and the coding length is determined according to the number of types in different categories.

3 Model construction

To dynamically select paths based on learners' real-time feedback, this paper adopts the Actor-Critic network architecture. The Actor part of this architecture fuses behavioral data and cognitive style data, processes the fused data through multiple fully connected layers to capture complex patterns in the state space, and outputs the probability of

Table 3 Partial sliding window calculation table

Time point t	Study time x_i	Sliding window average	Sliding window maximum value	Sliding window minimum
1	2.0	2.0	2.0	2.0
2	3.0	2.5	3.0	2.0
3	5.0	3.3	5.0	2.0
4	4.0	4.0	5.0	3.0
5	6.0	5.0	6.0	4.0

Table 4 One-Hot encoding table

Category	Type	One-Hot encoding
Learning preferences	Visual	[1,0,0]
	Auditory	[0,1,0]
	Kinesthetic	[0,0,1]
Information processing style	Analytical	[1,0]
	Intuitive	[0,1]
Way of thinking	Creativity	[0,1]
	Logical	[1,0]

selecting each learning path in the current state to select a learning path. In the Critic part, the current learning state and the selected learning path are evaluated, LSTM is used to process the relevant time series data, and the evaluated Q value (evaluation of the state-path pair) is output to dynamically adjust the learning path.

3.1 Actor part

The first level of the Actor part is the input layer, where the model receives the learner's behavioral data and cognitive style data. The behavioral data is represented as $x_b \in R^{d_b}$ and the cognitive style data is represented as $x_c \in R^C$.

The second layer consists of an embedding layer, which helps reduce data calculation complexity and enhances model representation capacity, this paper studies the use of the embedding layer to map discrete category data into dense vector representations of cognitive style data.

The mapping method is:

$$e_c = Embedding\ (x_c) \tag{7}$$

In formula (7), x_c is cognitive style data, and e_c is the dense vector representation of cognitive style data.

The third layer is the fusion layer. In this layer, this paper studies the weighted fusion of behavioral data and cognitive style data to generate a comprehensive representation of the learner state.

The weighting method is:

$$X_{fusion} = \omega_b \cdot x_b + \omega_c \cdot e_c \tag{8}$$

In formula (8), ω_b and ω_c are learnable weight coefficients. When the model is working, these two weight coefficients can be continuously updated according to learning, so as to achieve a fusion that is more in line with the current learning state.

The fourth layer is the MLP layer, and the input of this layer is the fusion data obtained by the fusion layer. There are four fully connected layers in the MLP layer designed in this paper according to Fig. 1, and each layer is activated by the nonlinear activation function ReLU.

ReLU activation formula:

$$ReLU(x) = \max\ (0, x) \tag{9}$$

The benefit of using the ReLU formula for activation is that it is simple and efficient, and since there are no negative numbers in the data studied in this paper, the "dead neuron" problem caused by negative inputs of ReLU is avoided.

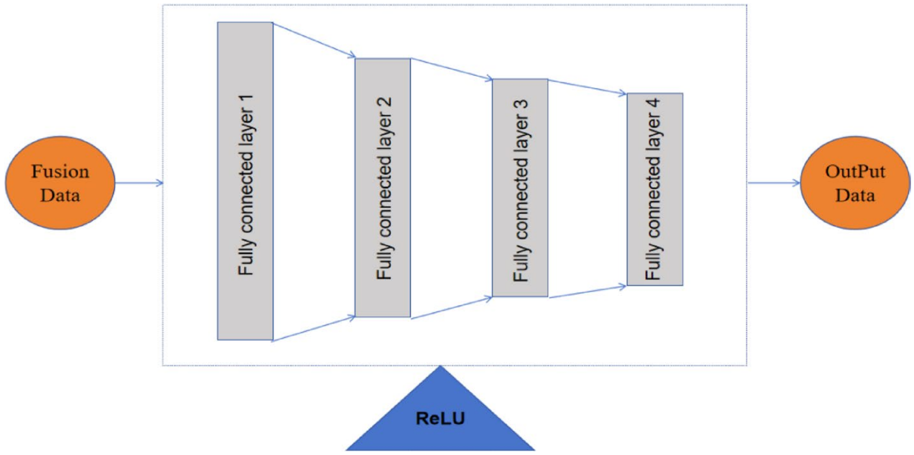

Fig. 1 MLP internal architecture

The last layer is the output layer of the Actor part, which is a Softmax layer. The main function is to output the probability of choosing a learning path in the current state, that is, if the Actor chooses a learning path a_t according to the current learner state s_t, then the probability of choosing the path is:

$$\pi(a_t|s_t) = softmax\,(W_a \cdot x_{fusion} + b_a) \tag{10}$$

In formula (10), W_a is the weight matrix of the actor, b_a is the bias term, and x_{fusion} is the fused data output.

3.2 Critic part

The first layer of the Critic is the input layer, which receives the current state data s_t of the learner and the current learning path a_t output by the actor, and concatenates the two to form a new input vector z_t:

$$z_t = concat\,(s_t, a_t) \tag{11}$$

The second layer is the LSTM layer and the internal structure of LSTM is shown in Fig. 2. Since the learning process is time-dependent, this paper uses the LSTM layer to extract time series data. This layer can remember past information and capture the impact of learning history on current decisions. The sequence input to the LSTM layer is the learner's state sequence and the corresponding learning path:

$$h_t = LSTM\,(z_t, h_{t-1}) \tag{12}$$

In formula (12), h_t is the hidden state of the LSTM layer, which combines the learner's current state learning path and previous learning history.

The third layer is the fully connected layer. This paper studies and designs three fully connected layers to further process the output data of the LSTM layer to generate Q value (evaluation index of state-path pair):

$$Q\,(s_t, a_t) = W_q \cdot h_t + b_q \tag{13}$$

In formula (13), W_q is the weight matrix, h_t is the output of LSTM, b_q is the bias term, and the final $Q(s_t, a_t)$ is the evaluation Q value of the current state-path pair.

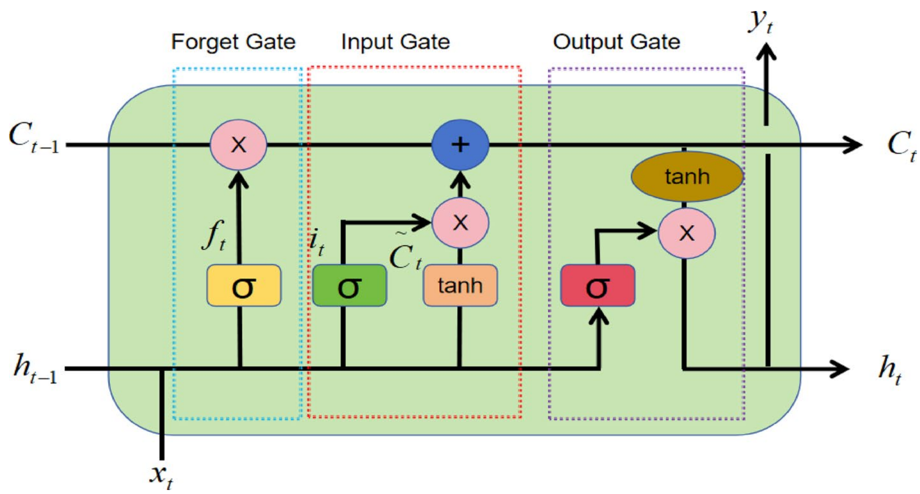

Fig. 2 LSTM internal structure diagram

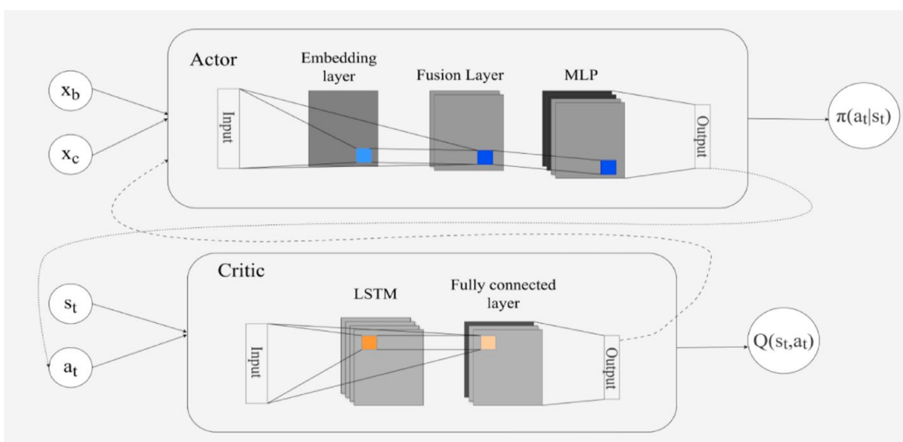

Fig. 3 Model architecture diagram

The last layer is the output layer. The Critic part uses the Q value to evaluate the value of the current learning path. The Q value represents the expectation of choosing this learning path under the current state. A scalar representing the Q value is output in this layer.

3.3 Model architecture diagram

As shown in Fig. 3, this is the architecture diagram of the model designed in this study. This architecture is composed of two primary elements: Actor and Critic. In the Actor part, the learner's behavior data and cognitive data are input into the input layer, and the cognitive data is mapped into a dense vector in the embedding layer and then fused with the behavior data in the third fusion layer into a comprehensive learner state data. The data enters the MLP layer and is processed by four fully connected layers. Finally, the probability of selecting each learning path in the learning state is output, so that the path is selected according to the maximum probability. In the Critic part, the current learning state and learning path are used as input, and after time series related processing in the LSTM layer, it is input into the fully connected layer for further processing and finally outputs a Q value that evaluates the current learning state and learning path. This value is returned to the Actor part as feedback to enter the next iteration.

3.4 Update strategy

In the Actor part, this paper uses policy gradient to update the strategy, the goal is to maximize the probability of learning path selection, the update formula is:

$$\Lambda_\theta J(\pi) = E_\pi[\Lambda_\theta \log \pi(a_t|s_t) \cdot A_t] \tag{14}$$

In formula (14), θ is the parameter set in the Actor part, and A_t is the advantage function, to determine the advantage of the current learning path, this evaluation method is employed. The calculation method is:

$$A_t = Q(s_t, a_t) - V(s_t) \tag{15}$$

In formula (15), $V(s_t)$ is the current state value function, representing the expected evaluation value under state s_t.

$V(s_t)$ is calculated as:

$$V(s_t) = W_v \cdot x_{fusion} + b_v \tag{16}$$

In formula (16), W_v is the weight matrix of the output layer, b_v is the bias term, and x_{fusion} is the output of the fusion layer.

The Actor partial update process based on policy gradient is shown in Fig. 4.

As shown in the flowchart in Fig. 4, in the Actor part, a path is first selected based on the current state, and then the policy gradient is updated according to the selected path.

In the Critic part, this paper uses the time difference error to update the Q value. The goal of the update is to make the Q value calculated by the Critic part as close as possible to the target Q value. The target Q value calculation formula is:

$$Q(s_{t+1}, a_{t+1}) = r_t + \gamma \cdot Q(s_t, a_t) \tag{17}$$

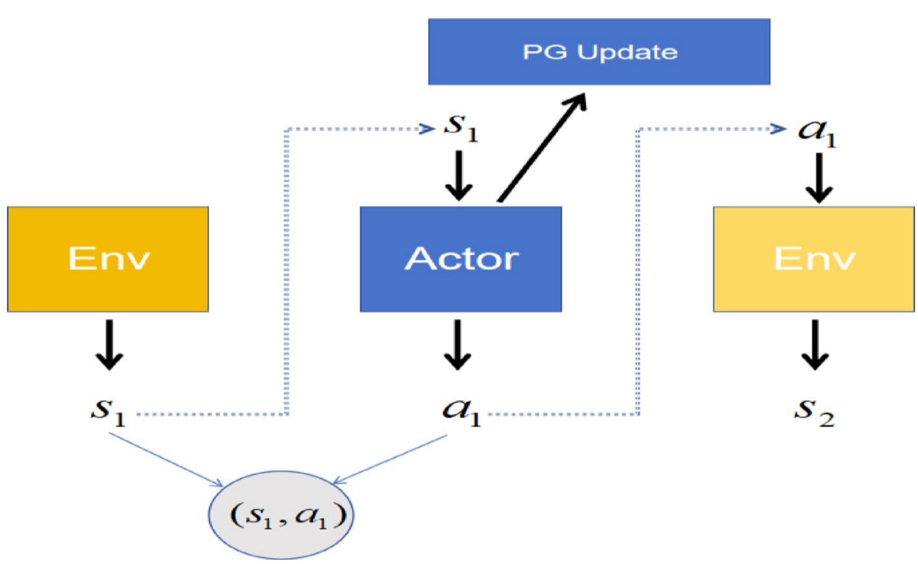

Fig. 4 Actor partial update flow chart

In formula (17), r_t is the immediate reward, which is calculated based on the feedback obtained after the learner takes the current learning path, and γ is the discount factor that controls the weight of future rewards.

r_t is calculated as:

$$r_t = f(\text{Learning progress, Study time, Test accuracy}) \tag{18}$$

The calculation formula for the update target of the Critic part is:

$$L(\theta_q) = E\left[\left(Q(s_{t+1}, a_{t+1}) - (r_t + \gamma \cdot Q(s_t, a_t))\right)^2\right] \tag{19}$$

The updated goal is achieved by minimizing the above function.

The process of updating the Q value of the Critic part based on the time-division error is shown in Fig. 5.

As shown in Fig. 5, the update of the Critic part first calculates the Q value, then calculates the time-division error according to the time-division error formula, and updates the Q value based on the calculated error.

4 Experimental design

The data used in the experiment include the learning behavior data, cognitive style data and historical learning record data of students in a certain school. A comparative experimental design is adopted, and the models compared include static DNN model, LSTM model and DQN model.

Multiple hardware and software devices are used in the experiment. In terms of hardware, the experimental equipment is equipped with Intel Xeon E5 series processors, NVDIA Tesla V100 graphics processing unit to accelerate the training process of deep learning models, and equipped with 32 GB memory and 1 TB SSD hard disk. In terms of software, Python is selected as the programming language, and deep learning frameworks such as TensorFlow 2.2 and PyTorch are used to build and train the personalized learning path optimization model implemented in this paper. Scikit-learn is mainly used for data analysis, which provides data analysis and evaluation functions for the model. In addition, the statistical analysis tools in the SciPy library are also used in the experiment to calculate P values and significance tests.

In the experiments of this paper, the ground truth is labeled based on the students' academic performance data, and the optimal learning mode is determined by analyzing the learning behavior path of students in the high-performance group. Specifically, we first divide students into high, medium, and low-performance groups based on their historical performance, and then extract the learning behavior characteristics of students in the high-performance group and use them as the reference standard for ground truth. This method is completely based on objective data, avoiding subjective bias, while

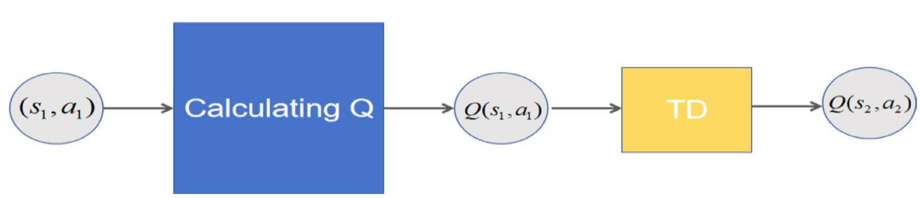

Fig. 5 Critic part Q value update flow chart

Table 5 Hyperparameter settings

Hyperparameters	Describe	Setting value
Learning rate	The learning rate controls the speed at which the model parameters are updated	0.01
Batch size	The number of data samples used in each training iteration	32
Optimizer	Optimization algorithm used for training	Adam
LSTM hidden layer size	The number of hidden units in the LSTM part	64
Dropout rate	The ratio of Dropout layers to prevent overfitting	0.2
Epochs	The total number of iterations of model training	100
Regularization coefficient	Coefficient used for L2 regularization	0.0005
Time steps	Time series step size used for LSTM network	10
Max path length	Maximum number of steps per learning path	50

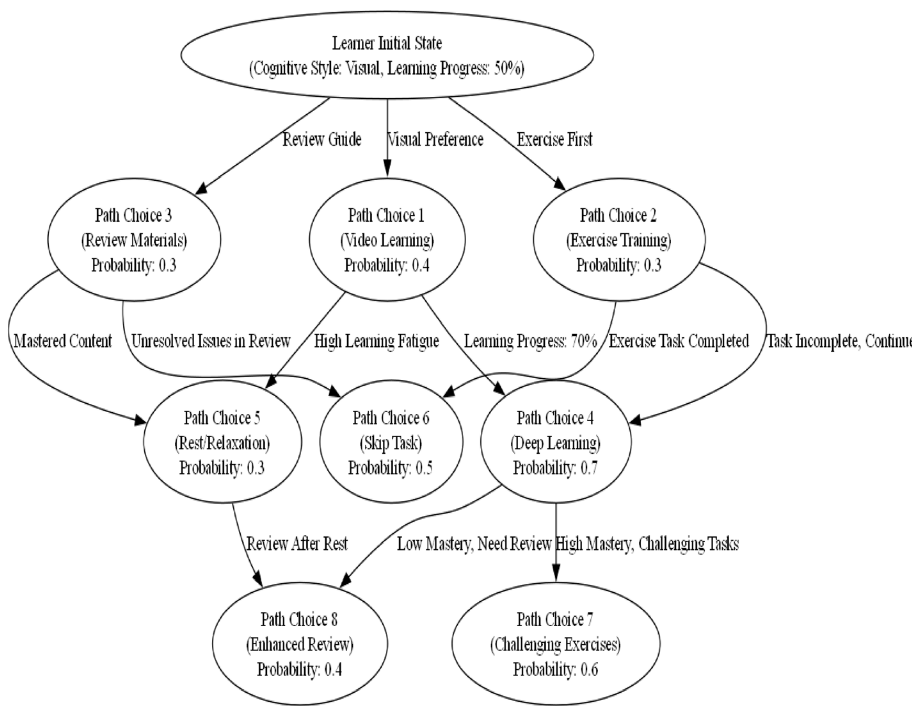

Fig. 6 Learning path analysis

ensuring the reproducibility and reliability of ground truth, providing a solid foundation for model training and evaluation.

The parameter settings for model training are shown in Table 5.

5 Result analysis

5.1 Learning path analysis

Figure 6 is a dynamic path selection display based on a learner from the initial state in this study. The initial state shows that the learner's cognitive style type is visual, and the current learning progress is 50%. The current learning state can determine the subsequent path selection. From the initial state down to the first level of path selection, there are three paths in total, namely video learning, exercise training and material review. Since the learner's cognitive style type is visual, the probability of video learning path being selected is higher, 40%, and the probability of exercise training and material review is 30% each. The learner's first-level path selection and the learner's state in the learning process (such as fatigue after learning for a period of time) can affect the subsequent path selection. For the second-level choices, if learners have a good grasp of the first-level video learning (learning progress ranges from 50 to 70%), they may proceed to in-depth video learning, with a probability of 70%. If learners feel tired after completing the first-level learning, they may choose to rest and relax, with a probability of 30%. At the same time, if the learner completes other tasks in the second layer except for the tasks selected in the first layer, he can choose to skip the second layer and enter the third layer of learning. In the third layer, learners can choose their final path according to their learning effectiveness evaluation. If the learner has mastered most of the learning content and the evaluation is good, he may directly choose to do challenging exercises to improve himself. If the mastery is not good, he may continue to conduct intensive review of the learning content.

5.2 Convergence analysis

Figure 7 is a convergence analysis diagram of the loss of the model studied in this paper during training. Training rounds are plotted along the horizontal axis, with loss value on the vertical axis. Since the model is an Actor-Critic model, this paper analyzes the loss of the Actor and the Critic separately. The blue line tracks variations in the Actor's loss value, and the red line shows variations in the Critic's loss value. The Actor and Critic loss curves drop rapidly in the early stage (0–40 rounds) and stabilize in the later stage (60–100 rounds), which shows that the update of model parameters during training effectively reduces the model error and the model can learn effectively and converge in the later stage. However, the final loss value of the model is close to 0, which indicates that the model may be at risk of overfitting.

Figure 8 compares the convergence of the loss value between the model studied in this paper and the traditional models Q-learning and DQN for the same task. For this task, the blue line reflects how the loss value of the model proposed in this study evolves, while the red and green lines correspond to the loss changes observed in Q-learning and DQN, respectively. Compared with the Q-learning and DQN models, the Actor-Critic

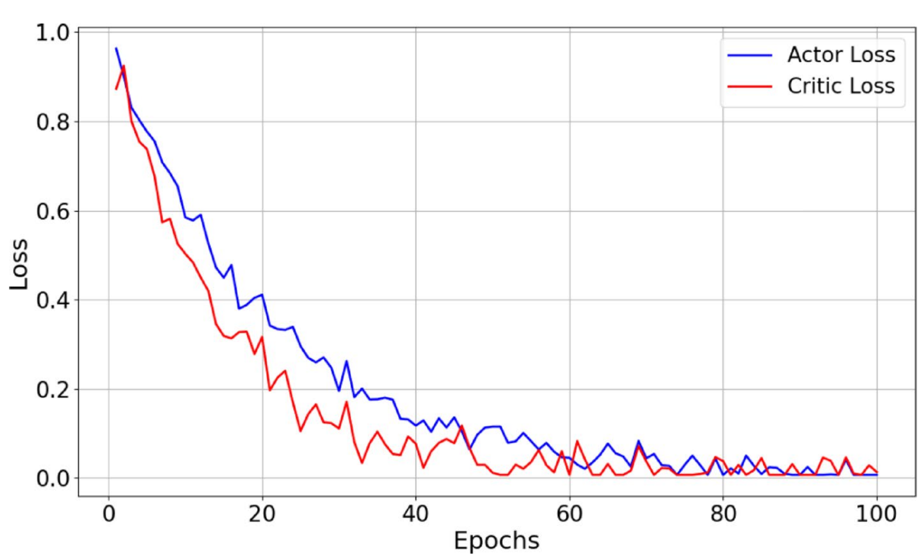

Fig. 7 Convergence analysis

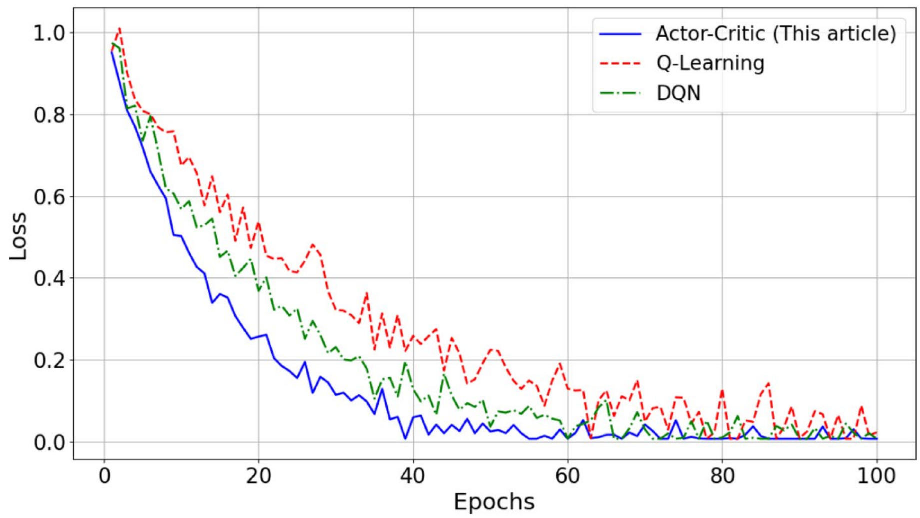

Fig. 8 Model error comparison

Table 6 is an analysis table of response time and average path adjustment calculation time

Group ID	Feedback response time (s)	Path adjustment computation time (s)
1	1.123	2.043
2	1.430	2.154
3	1.251	2.187
4	1.768	2.329
5	1.340	2.010

model studied in this paper has a significantly faster convergence speed and a smoother loss reduction process, indicating that the model has higher efficiency and stability in learning path optimization.

5.3 Response time and calculation time

Table 6 shows the response time and average path adjustment calculation time analysis of the model studied in this paper. Based on 500 experimental data, the experiments were divided into 5 groups, and the average feedback response time and path adjustment calculation time of each group were calculated. As can be seen from Table 5, the average feedback response time in the five groups of experiments is mostly concentrated between 1 and 2 s, while the average path adjustment calculation time is between 2 and 2.5 s. Among them, the shortest average feedback response time is 1.123 s, and the shortest average path adjustment calculation time is 2.010 s. These two shorter feedback times and path adjustment times significantly improve the real-time performance and application efficiency of the model. Especially in a dynamic learning environment, it can ensure a rapid response to changes in students' needs and timely adjust personalized learning paths.Experimental results demonstrate that the personalized path optimization model proposed in this study provides feedback to learners more quickly and generates new learning paths for dynamic adjustments.

5.4 Model comparison analysis

As shown in Fig. 9, it is a radar comparison chart of several performance indicators of the Actor-Critic model studied in this paper, the Q-learning model studied in traditional

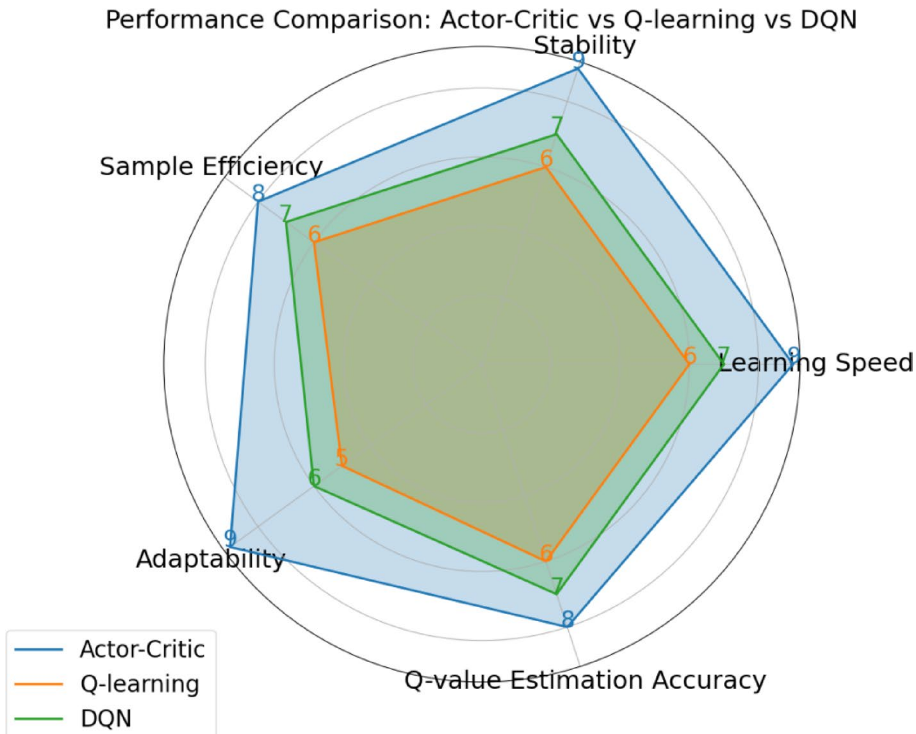

Fig. 9 Model comparison chart

research, and the DQN model. Five performances are compared, namely the model's learning speed, stability, sample efficiency, adaptability, and Q-value estimation certainty. By evaluating the performance of three different models in different aspects under the same task, the Actor-Critic model explored in this study outperforms traditional Q-learning and DQN models across five evaluated performance metrics, which shows the superiority of the model studied in this paper.

5.5 Effect of personalized learning path recommendation

To analyze the personalized learning path optimization effect of the model studied in this paper, the accuracy of personalized learning path recommendation was calculated and analyzed in three dimensions: learner's behavior data, cognitive style, and learning history, using the static DNN model, LSTM model, and DQN model.

In Fig. 10, the horizontal axis represents the three dimensions of personalized learning paths, and the vertical axis represents the accuracy of each model in different dimensions. Each network model has three bar graphs, representing behavioral data analysis, cognitive style analysis, and learning history analysis. The model studied in this paper has the highest accuracy in the personalized learning path recommendation in the three dimensions, which are 0.92, 0.90, and 0.93 respectively; the accuracy of the DQN model is 0.65, 0.50, and 0.60 respectively; the accuracy of the LSTM model is 0.75, 0.70, and 0.80 respectively; the static DNN model performs the worst, with accuracy of 0.55, 0.45, and 0.65 respectively. The statistical results of the three dimensions are compared, and the *P* values are calculated and displayed above each dimension. The final *P* values of the three dimensions are 0.02, 0.035, and 0.012, respectively, all less than 0.05, so it is

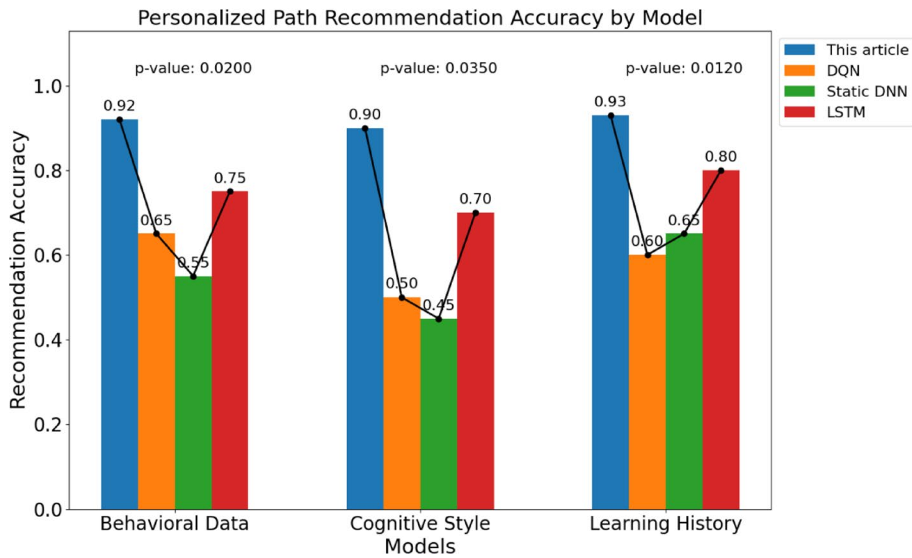

Fig. 10 Analysis of the accuracy of personalized learning path recommendation

Table 7 Ablation experiment results

Model configuration	Learning path recommendation accuracy	Iterations	Loss fluctuation	Calculation time
Baseline model (no Actor-Critic, no LSTM)	0.72	150	High	Longer
Actor-Critic only	0.79	120	Middle	Shorter
LSTM only	0.78	130	High	Shorter
Complete model (Actor-Critic + LSTM)	0.92	40	Low	Shortest

believed that the accuracy results of different models in these three dimensions are statistically different.

5.6 Ablation experiment

In order to evaluate the contribution of LSTM and Actor-Critic framework to the overall performance, an ablation experiment was conducted to systematically remove the contents of these two parts. The experimental results are shown in Table 7.

From the analysis of the results in Table 7, we can see that the model combining the Actor-Critic framework and LSTM is better than the model using either one or neither of them in all performance indicators. Among them, the LSTM model helps to improve the stability and accuracy when processing time series data, while the Actor-Critic framework optimizes the learning path selection and convergence speed of the model, proving the importance of combining the two for personalized learning path optimization.

6 Discussion

6.1 Strengths and innovations of the model

The personalized learning path optimization model (EDNN) proposed in this study demonstrates significant strengths and innovations in several aspects. Firstly, the model effectively integrates learners' behavioral data and cognitive styles by combining the Actor-Critic framework, which has often been overlooked in previous studies. By introducing a multilayer perceptron (MLP) in the Actor part to fuse these two types of data, the model can more accurately capture the personalized needs of learners. Secondly, the model employs a long short-term memory (LSTM) network in the Critic part to process time series data, enabling it to consider not only the current learning state but also the learners' historical learning behaviors, thus achieving more precise dynamic adjustments. Additionally, experimental results show that compared to Q-learning and DQN models, our model significantly outperforms in terms of learning speed, stability, and adaptability.

6.2 Limitations of the model and future work

Despite the excellent performance of the model in our experiments, there are some limitations. Firstly, the model's final loss value is close to zero, which may suggest a risk of overfitting, meaning the model may perform poorly when faced with new data. Secondly, the efficiency and scalability of the model when processing large datasets have not been fully validated. Future research can focus on the following areas: improving the model's generalization ability to reduce the risk of overfitting; optimizing the model's computational efficiency to handle larger datasets; and exploring the applicability and effectiveness of the model for different types of learners and various academic subjects.

6.3 Implications for educational practice

The findings of this study have important implications for educational practice. Firstly, the successful implementation of the model indicates that integrating learners' behavioral and cognitive style data through technological means can significantly enhance the quality of personalized learning path recommendations. For educational institutions, this means they can more effectively meet students' learning needs and improve teaching outcomes. Secondly, the model's rapid response and dynamic adjustment capabilities provide possibilities for real-time adjustments to teaching strategies, helping teachers to adjust teaching content and methods in a timely manner to accommodate students' learning progress and styles. Lastly, this study also provides new ideas for the development of educational technology, namely, by using deep learning and reinforcement learning techniques to develop more intelligent and personalized learning support systems.

6.4 Impact on learners

From the perspective of learners, the model proposed in this study can provide more personalized learning path recommendations, helping learners to plan and execute their study plans more effectively. With real-time feedback and dynamic adjustments, learners can understand their learning conditions in a timely manner and adjust their learning strategies based on feedback, thereby improving learning efficiency and outcomes. Additionally, personalized learning path recommendations also help to stimulate learners' interest and motivation, enhancing the autonomy and engagement in learning.

7 Conclusions

Due to the drawbacks of traditional research methodologies, this study designs a personalized learning path optimization model grounded in EDNN. The model combines the Actor-Critic framework, introduces MLP in the Actor part to integrate learners' behavioral data and cognitive style, and uses LSTM network to process time series data in the Critic part, facilitating the optimization and dynamic adjustment of personalized learning pathways. Experimental results show that the proposed model can effectively integrate learners' behavioral and cognitive characteristics, and achieve more accurate learning path selection and dynamic adjustment. Compared to traditional methods, the proposed model demonstrates significant improvements in learning speed, stability, and adaptability. However, the model convergence analysis shows that the final loss converges to near 0, suggesting that there may be a risk of overfitting, that is, the model may perform poorly when faced with new data. Therefore, future research can focus on improving the model's generalization ability, reducing the risk of overfitting, and thus improving its stability and effectiveness in practical applications.

Author contributions

Xiaomei Ding led the conceptualization and drafting of the manuscript, integrating behavioral and cognitive style data to optimize personalized learning paths. Huaibao Ding contributed to the data analysis and interpretation, particularly in the application of the Actor-Critic framework. Fei Zhou was instrumental in the technical development and implementation of the multilayer perceptron (MLP) and long short-term memory (LSTM) networks. Lihong Zhao, as the corresponding author, oversaw the project, ensuring the research met academic standards and was responsible for the overall direction and quality of the research.

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Data availability

The link to the dataset for this article is as follows: https://9jlp28t5.tcedshare.cn/web/share/5f35398e591a7dfdd61c6b8c3d71561b.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the ethical standards of the [Anhui Wenda University of Information Engineering] and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.The protocol was approved by the [Ethics Committee of Anhui Wenda College of Information Engineering] in accordance with the [Belmont Report, the Declaration of Helsinki].Informed consent was obtained from all individual participants included in the study. For participants under 18, informed consent was obtained from a parent and/or legal guardian with an assent from the participant.

Competing interests

The authors declare no competing interests.

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